

Design and Analysis of a Small Enterprise Network Topology



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1. Introduction

This report presents the design and analysis of a small enterprise network topology created using a network simulation environment. The topology demonstrates how different networking devices such as switches, firewall devices, servers, and personal computers can be connected together to provide secure communication and efficient data exchange.

The network includes two layer- 2 switches, one firewall device, one server, and two client computers. The design focuses on connectivity, segmentation, and secure communication between devices.

2. Objectives

The main objectives of this topology are:

- To build a functional enterprise network.
 - To demonstrate communication between hosts through switches.
 - To implement a firewall for network security.
 - To configure IP addressing for different devices.
 - To simulate real- world networking concepts.
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3. Network Devices Used

Device Name	Device Type	Purpose
FW2	Firewall (USG6000V)	Provides network security and traffic control
LSW1	Layer- 2 Switch (S3700)	Connects end devices and forwards frames
LSW2	Layer- 2 Switch (S3700)	Provides switching functionality for another segment
Server1	Server	Provides centralized services and resources
PC2	Personal Computer	Client workstation
PC3	Personal Computer	Client workstation

4. Network Topology Description

The topology consists of a firewall device placed at the center of the network. The firewall connects to two switches:

- LSW1 connects to PC3.
- LSW2 connects to PC2.
- The firewall also connects directly to the server.

This design allows communication between devices while maintaining centralized security through the firewall.

5. IP Addressing Information

PC2 Configuration

- IP Address: 10.1.2.100
- Subnet Mask: 255.255.255.0
- Default Gateway: 10.1.2.1

PC3 Configuration

- IP Address: 40.1.1.100
- Subnet Mask: 255.255.255.0
- Default Gateway: 40.1.1.1

Server1 Configuration

- IPv4 Address: 0.0.0.0
- IPv6 Address: 2000::2
- IPv6 Gateway: 2000::1

6. Connection Details

Source Device	Destination Device	Connection Type
FW2	LSW1	Copper Cable
FW2	LSW2	Copper Cable
FW2	Server1	Copper Cable
LSW1	PC3	Copper Cable
LSW2	PC2	Copper Cable

7. Firewall Role

The firewall device plays a major role in securing the network. Its responsibilities include:

- Filtering incoming and outgoing traffic.
- Preventing unauthorized access.
- Controlling communication between network segments.
- Enhancing network security.

The firewall acts as the gateway between the different network segments.

8. Switch Functionality

The switches are responsible for forwarding Ethernet frames based on MAC addresses. Their functions include:

- Connecting multiple devices within the LAN.
 - Reducing network collisions.
 - Improving communication efficiency.
 - Supporting network scalability.
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9. Advantages of the Topology

This topology provides several advantages:

- Improved network security using a firewall.
 - Organized network structure.
 - Easy scalability for future expansion.
 - Efficient communication between devices.
 - Better management of network traffic.
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10 . Conclusion

The implemented network topology successfully demonstrates the integration of switches, firewall devices, servers, and client computers within a secure enterprise environment. The topology provides efficient communication while maintaining centralized security and proper network organization.

This practical implementation helps in understanding important networking concepts such as switching, IP addressing, firewall protection, and enterprise network design.